Crop-Label Alignment: Auditing Crop-Dependent Auto-Labels in Remote-Sensing Segmentation Supplementary Material

Anonymous WACV Datasets Track submission

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Review artifact. The submitted code and compact evidence package are shared through an anonymous review repository. Public author-identifying URLs are withheld from this supplementary PDF during double-blind review.

1. Splits, units and what is held fixed

Sea ice. 169,051 patches of 128×128 pixels over 39 tiles and 17 acquisitions, each patch centred on an ICESat-2 token, drawn as a class-stratified subsample of 10,000 tokens per acquisition from three strong beams. Coverage is uneven by construction: the artefact set is a strip along the satellite track, and a source pixel is covered by between 1 and over 100 patches. Folds are leave-one-acquisition-out. Model selection uses the first two acquisitions in sorted order, which is the same pair for most folds and never includes the held-out one.

Floods. Sen1Floods11 [2] has 446 chips of 512×512 pixels over 11 events. The crop sweep takes 128×128 crops on a stride-32 grid, $13 \times 13 = 169$ positions per chip, drawing 24 per chip per epoch during training and evaluating κ on all 169. Folds are leave-one-event-out, model selection holds out a second whole event, and normalisation statistics are computed from that fold's training chips only.

The comparison. The primary conclusions use contrasts between arms with the same imagery, crop grid, architecture and sampling order under a fixed training seed. Raw arm values are also reported to expose the control floor. What varies is the training-label construction.

How the arms are built. Every arm is an explicit field of per-crop thresholds, one value for each of the 169 crop positions in each of the 446 chips, written to disk before any training. The dial arms interpolate between the published per-event constant and a predominantly per-crop Otsu field. Of 75,374 positions, 72,029 are direct crop fits (95.56%), 2,331 use a finite chip fallback (3.09%), and 1,014 in six chips remain unusable (1.35%). Arrays span all 446 chips; training, scoring and arm summaries use the common 440-chip set. The scramble arm randomly permutes threshold positions within each chip. The multiset of thresholds inside a chip is unchanged, which the builder asserts by sorting both and comparing. Over the full serialized 446-chip mapping (including the six excluded rows), 461 assignments (0.61%) are fixed points, 21.78% pair overlapping donor and recipient crops, and every donor shares the same chip. Mean threshold and within-chip spread are identical to the per-crop arm at $-17.796\,\mathrm{dB}$ and $1.453\,\mathrm{dB}$. The offset arm adds one $N(0, 1.453)$ draw per chip to the published constant, so it varies between chips at the same scale and not within them. The scramble attenuates but does not remove the threshold–input correspondence; the offset is crop-invariant within each chip.

Aggregation. In the flood arms, seeds are averaged within an event before anything else, because the event is the held-out evaluation unit. A held-out fold is not an independent replicate: the eleven models are leave-one-event-out folds sharing nearly all of their training data, which is why neither benchmark's fold summary carries a test.

Per-fold sea-ice results. Table 1 gives the sea-ice result one acquisition at a time, treated arm against its matched control on identical pixels, so the aggregate the main paper reports can be read against its parts.

acquisition	κ treated	κ control	difference	Ω tr.	Ω ct.	$\lvert\mathcal{A}\rvert$ %
20191103184432	+0.2019	+0.0051	+0.1968	0.0573	0.0185	7.4
20191104195311	+0.1568	+0.0025	+0.1544	0.0415	0.0171	13.1
20191106190152	+0.1952	+0.0118	+0.1833	0.0572	0.0344	12.1
20191111200211	+0.0413	+0.0000	+0.0413	0.1454	0.0000	4.0
20191113191053	+0.2257	+0.0155	+0.2102	0.0308	0.0227	7.5
20191115195353	+0.1105	+0.0296	+0.0809	0.0933	0.0640	23.1
20191116192813	+0.1759	+0.0052	+0.1707	0.0465	0.0256	11.5
20191117190234	+0.1218	+0.0066	+0.1152	0.0662	0.0396	20.4
20191118183654	+0.0955	+0.0139	+0.0816	0.1289	0.0861	27.0
20191119194532	+0.0632	+0.0141	+0.0491	0.0496	0.0446	14.1
20191120191952	+0.2216	+0.0031	+0.2185	0.0246	0.0114	5.9
20191121185413	+0.1436	+0.0052	+0.1384	0.0400	0.0226	10.5
20191122182834	+0.1327	+0.0079	+0.1249	0.0523	0.0182	12.1
20191123180255	+0.1934	+0.0104	+0.1830	0.0552	0.0398	9.5
20191124191133	+0.1812	+0.0201	+0.1611	0.0480	0.0265	7.7
20191126182014	+0.2301	+0.0101	+0.2201	0.0302	0.0173	7.3
20191127175434	+0.0903	+0.0048	+0.0855	0.1083	0.0461	31.6
mean	+0.1518	+0.0097	+0.1421	0.0632	0.0314	13.2

Table 1. Every sea-ice fold in the optical-only primary arm (seed 42). The treated model trains on the per-patch labels, the control on labels regenerated once per scene, both on identical pixels. The difference column is what the paper reports: mean +0.1421, descriptive sd 0.0585, positive in 17 of 17 dependent folds. $|\mathcal{A}|$ is the artefact set as a percentage of covered pixels. The photon-enabled seed-7 sensitivity is reported separately rather than averaged into this table.

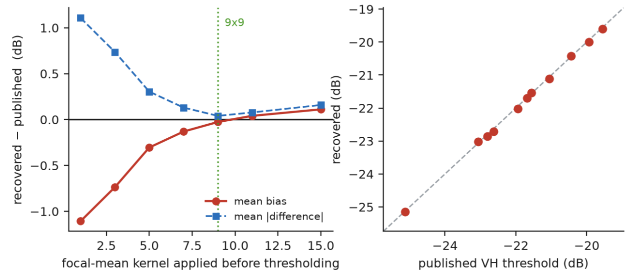

Figure 1. Left: bias and mean absolute difference against the published thresholds as the smoothing kernel is swept. Right: recovered against published threshold, one point per event. The bias crosses zero at a 9×9 focal mean, where 11 of 11 events land within 0.25 dB.

2. The full experimental configuration

Table 2 gives the complete configuration. Every cell is filled; "–" marks a setting that does not apply to that arm.

3. Limits of fold-level inference

Neither benchmark provides independent evaluation units for population inference. The 17 sea-ice folds are overpasses of 39 tiles across 25 days sharing geography and almost all training data; the eleven flood folds are leave-one-event-out models sharing nearly all of theirs. We therefore report effect sizes, descriptive standard deviations, per-fold values and training-seed sensitivity, but no p-values, confidence intervals, power calculations or required sample sizes over folds.

4. Association with expert-label performance

Sea ice has no reference labels, so it can show greater own-crop label alignment but not its relationship to human annotation. Sen1Floods11 has both. Training on each label set and scoring both against the expert labels on the same held-out event gives a transfer collapse of -0.0454 mIoU (sd 0.0470), negative in 9 of 11 events. Model input is Sentinel-1 throughout, since the expert labels were produced by analysts correcting a Sentinel-2 classification and the reference arm is open-loop only while the model never reads Sentinel-2.

The collapse is largest where labeller and human disagree most ($r = +0.719$), but this correlation does not identify a mechanism. An arm scored against expert labels is bounded by the agreement between the label sets, so that correlation is expected under any hypothesis, and dropping the two extreme events takes it to +0.32. This is an association between algorithmic training labels and expert-label performance. It does not isolate the contribution of crop dependence. We report it here because sea ice has no human reference.

This result is supplementary because it does not measure the crop-dependent artefact directly. Sea ice, where that artefact occurs, has no human reference; Sen1Floods11 supplies the reference needed to ask the performance question. The answer therefore applies to this algorithmic-label construction as a whole.

5. Recovering the flood labeller

Sen1Floods11 publishes its per-event threshold, so recovery of its labeller can be checked independently. Reading the raw band and thresholding it reproduces the published labels on 95.2% of pixels, but the recovered threshold sits 1.11 dB low on every one of the eleven events. The consistent sign points to a systematic cause: a smoothing step we had not applied. Figure 1 sweeps the radius: the bias crosses zero at a 9×9 focal mean, where all eleven events land within 0.25 dB at 99.08% agreement and the largest miss is 0.09 dB. That is the preprocessing the dial of the main paper holds fixed while it varies the threshold alone.

6. Stepwise ablation of the published labeller

The published labeller fits four quantities to whatever array it is handed, and Tab. 3 moves each of them to the scene in turn. Moving the median background and the Otsu threshold takes $P(\mathcal{A})$ from 37.04% to 4.50%. Moving either normalisation instead leaves the result bitwise identical.

	sea ice	floods, whole chip	floods, crop sweep
architecture	U-Net [8]	U-Net	U-Net
encoder	ResNet-18 [4]	ResNet-18	ResNet-18
encoder init	ImageNet [3]	ImageNet	ImageNet
optical channels	3	–	–
SAR channels	–	2 (VV, VH)	2 (VV, VH)
photon features	0 primary; 10 sensitivity	–	–
crop size	128×128	512×512 (whole chip)	128×128, stride 32
optimiser	AdamW [7]	AdamW	AdamW
learning rate	1e-4, cosine over the cap	3e-4, constant	3e-4, constant
weight decay	1e-4	1e-4	1e-4
loss	focal [6]	cross-entropy	cross-entropy
batch size	128	8	32
epoch cap	12	40	30
early-stop patience	5	10	8
held-out unit	acquisition	flood event	flood event
folds	17	11	11
seeds	42 primary; 7 sensitivity	42, 7, 123	42, 7, 123

Table 2. Configuration for the three experimental arms. Every cell is filled; "–" marks a setting that does not apply. Model selection is by best validation mIoU throughout. The sea-ice primary is seed 42 with the photon branch disabled; seed 7 is photon-enabled and is a separately reported input-mode sensitivity and is excluded from the primary average. The seven flood crop-sweep arms use seeds 7, 42 and 123; all three scramble models share one threshold reassignment.

fitted per crop				$P(\mathcal{A})$	
backgd.	Otsu	norm. 1	norm. 2	mean	max
•	•	•	•	37.04%	57.90%
	•	•	•	19.46%	40.85%
•		•	•	26.93%	47.71%
•	•		•	37.04%	57.90%
•	•	•		37.04%	57.90%
		•	•	4.50%	40.07%
				0.00%	0.00%

Table 3. The cloud and shadow stage of the released labeller, run per crop on 446 public Sen1Floods11 optical chips, 169 crops each, nothing trained. A dot marks a quantity estimated from the crop rather than from the whole scene, so the top row is the pipeline as published. Bands are stacked in the order cv2.imread produces, which is what the pipeline is fed. Stage 1 alone, which fits nothing, gives 0.00% on every chip.

A min-max is the identity on an array that already spans the range it maps onto, and with the background and the Otsu threshold fitted per crop that is what it is handed. The array reaching the first normalisation already spans 0 to 255 in 98.7% of the 75,374 crops, and the array reaching the second already spans the scene's own second range in 99.0%. On the 787 crops where the two ranges genuinely differ, so the substitution has something to do, we compared the class maps pixel by pixel and found no pixel changed, for either normalisation. Constant arrays, where the two branches could diverge because OpenCV maps a constant to zero, occur in 956 crops and change nothing either.

Repair the background and the Otsu threshold and the array is narrower and varies between crops, and the normalisations then carry the 4.50% that is left. A step can be inert beside a larger one and become the whole of what remains once that one is fixed.

How much of this is our preparation. Reflectance is scaled to 8 bits by one global constant, so no per-crop statistic enters before the rule under test. Thus, a nonzero $P(\mathcal{A})$ originates in the rule under test. Its magnitude can still depend on the constant, so we varied that constant. At ceilings of 2000, 3000 and 5000, which clip 11.5%, 3.5% and 0.8% of samples, the published pipeline reads 35.04%, 37.04% and 37.48%, and the residual after repairing the background and the Otsu threshold reads 2.99%, 4.50% and 6.18%. The headline is stable and the residual is not, moving by a factor of two across a range of ceilings that changes the clipped fraction fifteenfold. It stays clearly nonzero throughout, which is the claim it is asked to carry, and it is not a quantity to compare across papers.

Repairing the most obvious step may leave an artefact in a later step, and the order of repairs can change which component appears responsible. The full ablation exposes that interaction more clearly than a single before-and-after comparison.

7. A real-image example from the published labeller

Figure 1 of the main paper is a schematic, so Fig. 2 is the same picture built from a Sentinel-2 scene by shadow_cloud, the published function, in the two modes the ablation above uses. We train no model and introduce no learned parameters beyond quantities fitted by that routine.

The scan is reproducible as printed: fig_mechanism_real.py –scan with the shipped defaults proposes ten sites in each of twelve scenes at seed 0, rejects windows crossing swath boundaries, accepts 61, and writes mechanism_survey.json. The four numbers above are recomputed from that file. A single region demonstrates the mechanism; the scan characterises randomly proposed sites, while the displayed panel was selected descriptively from the accepted scan. Sites are drawn uniformly at random from twelve scenes, rejected only if the window falls off the swath, and each is measured on the overlap of the same two crops. That the median is near 1% while the tail reaches 39% is the expected shape: the stage fits a threshold to whatever is in the window, so a window of uniform ice gives the two crops nothing to disagree about, and a window split by a cloud edge gives them a great deal.

8. Why a sparser grid reads a larger alignment

Thinning the evaluation grid from stride 32 to stride 64, on identical models and labels with nothing recomputed, takes the artefact set from 24.71% of covered pixels to 16.31% and the contrast from +0.2666 to +0.4587. The sparser grid reads larger for two reasons that pull the same way: the artefact set shrinks to the pixels whose covering crops disagree most strongly, and the second term of the estimator averages over fewer and more distant crops. What survives is the ranking, at Spearman $r = +0.918$ over the eleven events. This is why a κ reported without its crop size and stride cannot be interpreted, and why every comparison in the main paper is made within one geometry.

9. Encoder sensitivity

Every primary-table and primary-figure κ in the main paper comes from a U-Net with a ResNet-18 encoder; the robustness paragraph also reports two alternative encoders. The main paper explains why a trained network can sit above the exact null in terms of zero padding and receptive fields truncated at the crop border. Those properties create a possible confound. If κ mainly reads the architecture, every number in this paper would need to be scoped to one encoder.

So we reran the two ends of the dial on all eleven events with two other encoders, a Mix Transformer and a second convolutional network. We held the labels, folds, seed, schedule, epochs, artefact set and scorer fixed. Before launching we fixed what would falsify the reading: the $\alpha = 1$ minus $\alpha = 0$ contrast collapsing toward zero.

It did not, on either. The transformer reads a contrast of +0.2966 and the second convolutional encoder +0.2943, both positive in 11 of 11 events, against +0.2666 for the original U-Net on the same seed. All three land within 0.03 of each other, and the trained-control floor remains empirically near zero in all three. Ω rises from about 0.013 to about 0.113 between the arms on both replacements, so each becomes crop-dependent when its labels are.

This experiment does not show that κ is blind to padding. A Mix Transformer is not padding-free: its overlapping patch embedding is a strided convolution with padding, and here it drives a convolutional decoder. What the experiment supports is narrower and still worth having: the reading is not specific to the encoder it was measured on, and the magnitude moves by about 10% across the three. We did not test for equivalence. Whether an architecture genuinely without padded convolutions would read the same is untested.

10. Foundation-model mask survey protocol

The SAM survey uses the ViT-B model of Kirillov et al. [5] with checkpoint filename sam_vit_b_01ec64.pth and automatic-mask generation at 16 points per side, prediction-IoU threshold 0.85 and stability threshold 0.90. The first 37 valid Sen1Floods11 chips in sorted order are converted from VH to three identical 8-bit channels with one fixed $[-30, 0]$ dB range. Masks are generated either once on the 512×512 chip or separately on each 128×128, stride-32 crop. Each mask receives water/land from its mean VH at one fixed -17.5 dB threshold; unclaimed pixels use the same threshold pointwise. The stored result contains per-chip $P(\mathcal{A})$ values, but not the checkpoint hash, so the release supports reaggregation rather than byte-level reconstruction of SAM inference. A separate 24-pair repeat records identical masks and labels on every pair under the environment used here. This duplicates SAR into RGB channels and is an out-of-distribution stress test. It does not estimate the prevalence of SAM-based auto-labelling practice.

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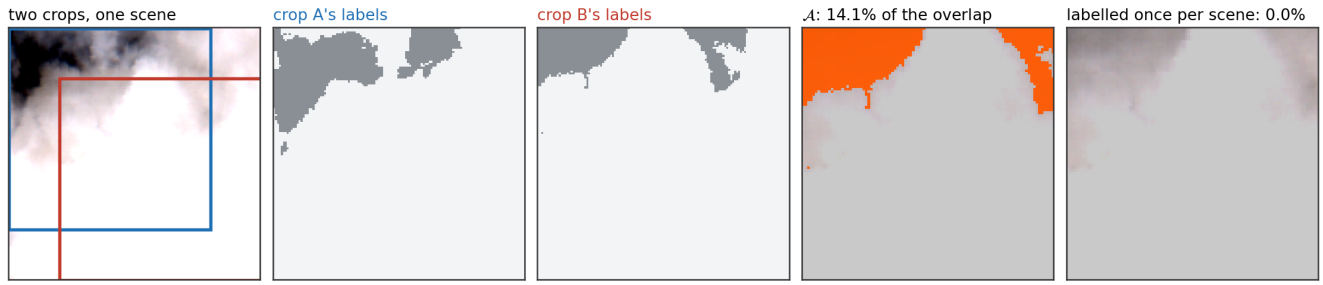

Figure 2. Two 128×128 crops 32 pixels apart on one Sentinel-2 region, labelled by the published cloud and shadow stage. The two crops see the same cloud edge against different windows, so the Otsu threshold each fits differs and the same source pixels come out ice in one crop and thin ice in the other: $\mathcal{A}$ is 14.1% of the overlap and traces the cloud boundary. Fitting the same four quantities once over the region instead makes the two crops agree everywhere, and $\mathcal{A}$ is empty. The region was chosen from a scan of 61 sites over 12 scenes, reported below: 53 of the 61 have a nonempty artefact set, the median is 1.23% and the largest is 38.90%, so this example sits above the median and well below the worst. Contrast is stretched between the 2nd and 98th percentiles of the region for display, which changes no number here.

rule	fit inside crop?	mean $P(\mathcal{A})$	bal. acc.
fixed B02 constant	no	0.00%	0.560
fixed whiteness + brightness	no	0.00%	0.591
clear-sky percentile, scene	no	0.00%	0.641
clear-sky percentile, crop	yes	26.57%	0.562
Otsu on haze transform, crop	yes	35.37%	0.590
min-max per crop, then fixed	yes	28.12%	0.600

Table 4. Six illustrative cloud-masking rules on 342 public CloudSEN12+ [1] patches, 509×509, 144 crops each, nothing trained. Values are unweighted means over patches. Balanced accuracy is against the expert mask, after majority vote over covering crops (ties go to clear sky). Rows three and four are the same estimator on different support.

11. Cloud-mask survey with expert labels

The rules in Tab. 4 are intentionally simple illustrative families; they do not reimplement a named operational product. They cover fixed reflectance, fixed whiteness and brightness, a clear-sky percentile, Otsu on a haze transform, and cropwise min–max normalisation. The percentile matters because it is a fitted quantity: run per scene it is one constant and $P(\mathcal{A})$ is zero; run per crop it becomes the hazard.

The accuracy column exists because $P(\mathcal{A})$ cannot see a semantic error. It is invariant to swapping the classes, to thresholding the wrong band, and to whether the rule works at all, so a control built from $P(\mathcal{A})$ alone cannot fail on anything but bookkeeping. CloudSEN12+ is the only dataset in these surveys that ships expert labels beside the imagery, which is what makes the column possible.

It reports balanced accuracy on the labels each rule actually produced. A scene-wide application would score a different rule. Both choices were forced. Intersection over union looks like the natural measure and is useless here: its value for a rule that calls every pixel cloud is the cloud fraction itself, about 0.46 on these patches, which beats every real rule in the table and would have certified the degenerate labeller the column exists to catch. Balanced accuracy is 0.5 for any constant rule and below 0.5 for an inverted one. All six rules clear it. And scoring a crop-fitted rule by applying it once to the whole scene measures a different rule from the one that produced the $P(\mathcal{A})$ beside it, which would make rows three and four identical by construction rather than by measurement.

They are not identical. Fitting the same clear-sky percentile inside each crop rather than over the scene is associated with 0.079 lower balanced accuracy, 0.562 against 0.641, while $P(\mathcal{A})$ moves from zero to 26.57%. This is a descriptive paired-rule comparison on a dataset we did not assemble.

Dependence on scene content. If $P(\mathcal{A})$ were a consequence of a rule being fitted and nothing more, it would not vary with what the imagery contains. Binned by the expert cloud fraction it rises monotonically for all three fitted rules: over patches that are under a tenth cloud it averages 19.6 to 24.9%, and over patches more than seven tenths cloud, 30.0 to 40.5%. It is highest on the most overcast patches rather than on the half-cloudy ones where cloud boundary is longest, which is what a fitted threshold should do: on a nearly uniform crop the quantile it lands on is arbitrary, and neighbouring crops make different arbitrary choices.

Scope. $P(\mathcal{A}) = 0$ means the labeller gives each source pixel one label, so there is nothing to detect and nothing to average over: κ is undefined there rather than zero, and our implementation raises on it. The converse does not hold: a large $P(\mathcal{A})$ places no lower bound on κ, and no claim here should be read as one. $P(\mathcal{A})$ is also defined relative to a grid and to a fitting support, and it is zero exactly when the support contains every crop covering a pixel. Fitting per scene achieves that for this grid; a patch is 5.09 km and a Sentinel-2 tile is far larger, so a coarser grid over the full tile would find the patch-level threshold crop-dependent one level up. The recommendation is to make the fitting support contain the grid, not that any particular support is physically correct.

12. What is released, and how to obtain the data

The estimator is one file, cropalign.py, requiring numpy alone, verified in a clean environment containing nothing else. It exposes a single class used in two passes over the crops, carries the six self-test controls described above, and raises on an empty artefact set. Every survey script, every result file it reads, and the number checker that audits this paper against those files are released with it.

All imagery is public. Sen1Floods11 is 1.6 GB from a Google Cloud bucket and needs no credentials; it carries the calibration, the expert-label comparison, the labeller recovery and both flood surveys, which is most of this paper. CloudSEN12+ is CC0, and the subset used here is a 1.84 GB archive of 342 GeoTIFF patches whose repository revision and SHA-256 are recorded in the released results, so the numbers can be tied to the bytes that produced them. The sea-ice half additionally needs Sentinel-2 L1C tiles from Copernicus and 72 GB of ICESat-2 ATL03 granules from NSIDC, the latter requiring an Earthdata account; no credential appears in the release.

The 68 sea-ice primary and input-mode-sensitivity runs were not metered separately, so we do not claim a total sea-ice training cost. The whole-chip flood models used about 3 GPU-hours and the J1 crop sweep 15.4. These are aggregate GPU-hours, not two-card wall time; inference audits are itemised separately in REPRODUCE.md. Three of the four label surveys need no GPU at all and run on CPU in minutes; only the one that calls a segmentation model does. Auditing the primary flood and sea-ice alignment tables without rerunning inference needs only the stored per-run scorer outputs: the verifier reaggregates their exact design cells and fails if one is missing. Results for which only summaries were retained are explicitly classified as re-read rather than recomputed.

Provenance, licensing and intended use. The sea-ice benchmark is ours: 169,051 patches of 128×128 pixels cut from Sentinel-2 L1C over 39 tiles and 17 acquisitions between 3 and 27 November 2019 in the Ross Sea sector, each centred on an ICESat-2 ATL03 token, split leave-one-acquisition-out. It is an assembly of public sources, not a new collection, and the assembly choices are ours: the co-location rule, the patch geometry, the class-stratified subsample. The flood arms are Sen1Floods11 as published, 446 chips over 11 events, split leave-one-event-out, with the per-crop threshold fields we build on top of it. CloudSEN12+ supplies 342 patches for the cloud survey.

What the archive actually contains. The attached anonymous archive is built by an allow-list: the estimator and its tests, analysis and survey scripts, stored result files, the claim-bearing LaTeX files scanned by the checks, both submitted PDFs, the exact bytes and hashes of 68 historical sea-ice training records, verification scripts, REPRODUCE.md and the licence. It is not a compile-ready copy of the conference template. The manifest records every payload path, size and SHA-256. It does not contain imagery, model checkpoints, the per-crop threshold fields, or per-run human-label damage scores.

The manifest names five verification tiers. The estimator's controls and lifecycle regressions run in the archive and pin its behaviour. The primary flood and sea-ice κ tables are reaggregated from exact per-run design cells, one file per scored model. Survey, architecture and grid summaries are checked as stored-value consistency claims rather than reconstructed from raw imagery. The accuracy loss and the 94% scramble figure are re-read from per-event summaries and not rebuilt, because no per-run damage scores were retained; that is a gap in what the archive can prove, not a claim about what is true. Scoring model predictions are not reproduced, since they need the checkpoints and the imagery. Training is not reproduced and is documented separately in REPRODUCE.md.

As a separate implementation check, the dependency-light released estimator and the training pipeline's scorer, which share no aggregation code, were run on the same retained flood checkpoint, labels and crop grid. They agreed on both weightings and every reported component to a worst absolute difference of 5.6×10^{-17}. The comparison script ships; rerunning it requires the omitted checkpoint and imagery. The archive therefore records this as a performed cross-check that it cannot reproduce.

The historical sea-ice training JSONs predate complete optimiser and data fingerprints. Their exact

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bytes and SHA-256 values ship in one anonymous payload; the verifier checks the complete design and derives input mode from their recorded photon field. They store the arm, acquisition, seed, input mode, encoder and loss; the remaining settings in Tab. 2 are reconstructed from the frozen launcher rather than independently authenticated by those JSONs. The current launcher fails closed on such legacy caches: fresh runs record the full optimiser/schedule, batch, AMP and stopping configuration plus a content manifest over code, imagery, arm-specific labels, the token table and feature schema. Fresh scores are bound to their checkpoint and training metadata, scorer/model code, evaluation labels and scene-valid masks. The standalone live-summary command enforces those bindings; the archive verifier separately checks the disclosed legacy outputs.

A generated MANIFEST.json carries a SHA-256 for every shipped payload file and separately records the hashes of the three scorers. Every scorer output records both weightings. Every primary J1 flood model-scorer output records norm_source=fold-local; four threshold-rule controls have no model normalisation. The scorer reads the numeric optical moments from each training run, whose metadata is not shipped. The sea-ice primary (seed 42) is optical-only and uses fixed ImageNet RGB constants. The sensitivity (seed 7) also uses those optical constants and photon-feature moments computed over every non-test acquisition in the fold (including the validation acquisitions). The released API names the source-pixel value kappa (also kappa_pixel) and the secondary value kappa_crop_read; legacy pipeline JSON instead uses kappa_pixel for source-pixel and kappa for crop-read. The manifest records this crosswalk explicitly. DESIGN.json carries the seventeen acquisition identifiers and the eleven event names, so the completeness gate compares the scored set against a frozen design. A substituted identifier, a duplicated cell and an unlisted file each fail it, and each was planted to confirm that. ARCHIVE_ID.txt holds one SHA-256 over the sorted path-and-hash list of the payload, which identifies that payload without naming us. We do not print that digest here: this document is inside the archive, so a digest quoted in it could never match the archive it describes. The builder refuses to package an archive in which its text and PDF scans find a configured identity or repository URL; the scan is checked against planted examples before it runs. The code is MIT and covers our own files only; the imagery keeps its source's licence. CloudSEN12+ is CC0 1.0. Sentinel-1 and Sentinel-2 are Copernicus Sentinel data under the Legal Notice on the Use of Copernicus Sentinel Data and Service Information. ICESat-2 ATL03 is distributed by NSIDC as a NASA Earth science data product, in the public domain and free to use with attribution. Sen1Floods11 is released by its authors under CC BY 4.0.

Intended use. The intended use is auditing a segmentation benchmark for crop-dependent labelling, on a stated crop grid, and comparing arms measured on that same grid. It is not a dataset for training a sea-ice product, not a benchmark leaderboard, and not a source of operational ice charts: the labels are algorithmic throughout, the sea-ice half has no human reference at all, and the paper's own subject is that such labels can be wrong in a way that a model will happily reproduce.

13. Both weightings of κ

The main paper reports κ weighted uniformly over source pixels. Weighting over (pixel, crop) instances instead gives κ_{cr}, in which a pixel counts once per covering crop. Every scored run records both.

	source-pixel	crop-read
flood, $\alpha = 1$	+0.2605	+0.2468
flood, $\alpha = 0$	−0.0008	−0.0008
flood, permutation	+0.0672	+0.0653
flood, end-to-end contrast	+0.2613	+0.2477
sea ice, treated (primary)	+0.1518	+0.1260
sea ice, control (primary)	+0.0097	+0.0154
sea ice, difference (primary)	+0.1421	+0.1106

The flood grid covers every interior pixel sixteen times, so the two weightings barely separate there. On $\mathcal{A}$, sea-ice coverage runs from 2 to over 100, and there the crop-read weighting understates the effect and overstates the floor: densely covered pixels sit along the satellite track, where the control's own crop-dependence is largest.

14. Reproducibility and tools

The released verifier runs six estimator controls, eighteen lifecycle regressions, reaggregates the primary flood and sea-ice κ tables from per-run outputs, checks the exact 17-acquisition and 11-event design products, and validates every listed payload hash. It reports summary-only accuracy quantities as re-read, not reproduced; imagery, checkpoints, training and model inference are not shipped. An AI assistant was used for code auditing, debugging, experiment execution and drafting. The authors designed the study and are responsible for every claim, number and omission.

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